

Electricity Demand Forecasting Using AI

Project Lab Automation

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Explained model

Open tasks

- Incorporate data from the equivalent date in previous year(s)
- Accuracy over added feature analysis

1. Create a dataset ☒
2. Graphical exploration ☒
3. Comparing and choosing the best model ☒
4. Training and backtesting ☒
5. Exogenous Variables ☒
6. Feature selection (explanability and interpretability) ☒
7. Iterate through 3.-6. until satisfied with the results ☒

Seasonal features

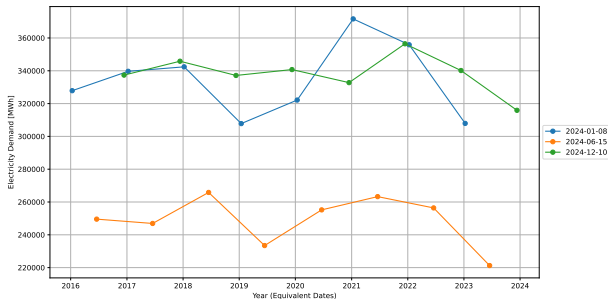


Figure 1: Seasonal features.

Electricity demand at weekday-aligned equivalent dates shows high variability across years and reference dates

→ Limited additional predictive value of these seasonal features

Problem:

- Included power generation data which are only available with one hour delay
- We predict 24 hours, hence this data are not available at the moment

Solution:

- Take the forecast instead of the real data which are also available at the SMARD-API [1]

Reminder: Feature Importance

Two complementary measures are considered:

- **Split importance:** number of times a feature is used for splitting
- **Gain importance:** total reduction of the loss function achieved by splits using the feature

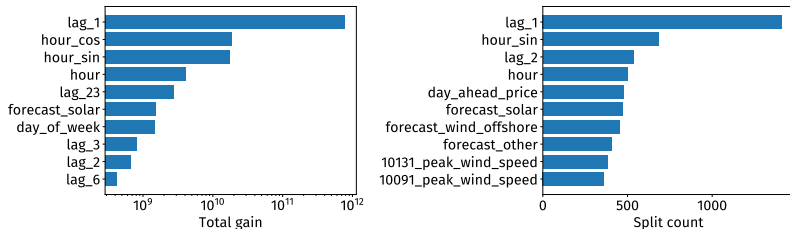
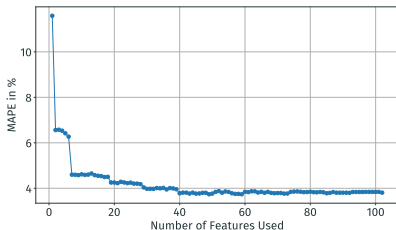
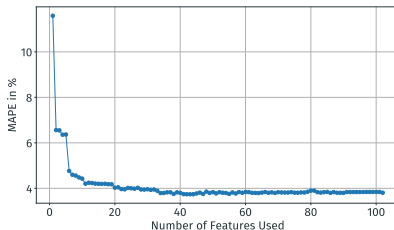


Figure 2: Top 10 features with respective importances

Feature Selection



(a) Total Gain.



(b) Split count.

Figure 3: MAPE as a function of the number of selected features using rank-based feature addition.

Gain gives faster convergence than split count

Notice: no insights about which feature combination gives optimal results

Should we include more model explanation, i.e. Shap Values?



Bundesnetzagentur.

Smard – strommarktdaten für deutschland.

<https://www.smard.de/home>, 2025.

Zugriff am 13. November 2025.